

## Deriving from sine-cosine identities

### 🔥 Memorize

$$\cos^2 \theta + \sin^2 \theta = 1$$

$$\sin(2\theta) = 2 \sin \theta \cos \theta$$

$$\cos(2\theta) = \cos^2 \theta - \sin^2 \theta$$

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

Using these identities, one can derive other identities and techniques when they are needed. It is important to practice deriving the other identities and techniques so that they come quickly in the moment.

### ⚠ Practice these derivations!

(1) **Derive**  $1 + \tan^2 \theta = \sec^2 \theta$

*Method:* Divide by  $\cos^2 \theta$ :

$$\cos^2 \theta + \sin^2 \theta = 1 \quad \ggg \quad \frac{\cos^2 \theta}{\cos^2 \theta} + \frac{\sin^2 \theta}{\cos^2 \theta} = \frac{1}{\cos^2 \theta}$$

$$\ggg \quad 1 + \tan^2 \theta = \sec^2 \theta$$

(2) **Derive**  $\cot^2 \theta + 1 = \csc^2 \theta$

*Method:* Divide by  $\sin^2 \theta$ :

$$\cos^2 \theta + \sin^2 \theta = 1 \quad \ggg \quad \frac{\cos^2 \theta}{\sin^2 \theta} + \frac{\sin^2 \theta}{\sin^2 \theta} = \frac{1}{\sin^2 \theta}$$

$$\ggg \quad \cot^2 \theta + 1 = \csc^2 \theta$$

(3) **Derive**  $\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$

*Method:* Use  $\tan \theta = \frac{\sin \theta}{\cos \theta}$ :

$$\begin{aligned}\tan(2\theta) &= \frac{\sin(2\theta)}{\cos(2\theta)} \ggg \frac{2 \sin \theta \cos \theta}{\cos^2 \theta - \sin^2 \theta} \\ &\times \frac{1}{\cos^2 \theta} \ggg \frac{2 \tan \theta}{1 - \tan^2 \theta}\end{aligned}$$

(4) **Derive** “power to frequency conversion”:

$$\sin^2 \theta = \frac{1}{2}(1 - \cos(2\theta))$$

$$\cos^2 \theta = \frac{1}{2}(1 + \cos(2\theta))$$

*Method:* Solve the  $\cos(2\theta)$  identity for  $\sin^2 \theta$  or  $\cos^2 \theta$ .

### Extra - Deriving other identities from the sine sum identity

In fact, assuming the relations like  $\cos \theta = \sin(\theta + \frac{\pi}{2})$ , one can derive the other trig identities from the Pythagorean identity and the sine sum-angle identity.

For example:

$$\sin(2\theta) = \sin(\theta + \theta) \ggg \sin \theta \cos \theta + \cos \theta \sin \theta \ggg 2 \sin \theta \cos \theta$$

Another example:

$$\cos(2\theta) = \sin(\theta + (\theta + \pi/2)) \ggg \sin \theta \cos(\theta + \pi/2) + \cos \theta \sin(\theta + \pi/2)$$

$$\ggg \sin \theta(-\sin \theta) + \cos \theta \cos \theta$$

$$\ggg \cos^2 \theta - \sin^2 \theta$$

Another example:

$$\cos(A + B) = \sin(A + (B + \pi/2)) \ggg \sin A \cos(B + \pi/2) + \cos A \sin(B + \pi/2)$$

$$\ggg \sin A(-\sin B) + \cos A \cos B$$

$$\ggg \cos A \cos B - \sin A \sin B$$

## Flip and rotate identities

There are many minor trig identities involving angles  $-\theta$  or  $\theta \pm \pi$  or  $\theta \pm \pi/2$ .

### Flip

To handle  $\text{Trig}(-\theta)$ , use these two basic identities along with the *definitions* of other trig functions as needed:

### 🔥 Memorize

$$\begin{aligned}\sin(-\theta) &= -\sin \theta \\ \cos(-\theta) &= +\cos \theta\end{aligned}$$

### ≡ Example

$$\begin{aligned}\cot(-\theta) &= \frac{1}{\tan(-\theta)} \quad \ggg \quad \frac{\cos(-\theta)}{\sin(-\theta)} \\ &\ggg \quad \frac{+\cos \theta}{-\sin \theta} \\ &\ggg \quad -\cot \theta\end{aligned}$$

## Quarter-turn rotation

Angles  $\theta \pm \pi/2$  actually convert between sine and cosine, potentially with  $\pm 1$  factors:

### 🔥 Memorize

$$\begin{aligned}\sin(\theta + \pi/2) &= +\cos \theta \\ \cos(\theta + \pi/2) &= -\sin \theta\end{aligned}$$

For  $\theta - \pi/2$ , just rewrite it as  $-(-\theta + \pi/2)$  and apply a negation identity first to peel off the outer minus.

For other trig functions, apply the definitions.

### ≡ Examples

Find  $\sin(\theta - \pi/2)$  like this:

$$\begin{aligned}\sin(\theta - \pi/2) &\ggg \sin(-(-\theta + \pi/2)) \\ &\ggg -\sin(-\theta + \pi/2) \\ &\ggg -\cos(-\theta) \\ &\ggg -\cos \theta\end{aligned}$$


---

Find  $\csc(\theta - \pi/2)$  like this:

$$\begin{aligned}\csc(\theta - \pi/2) &\ggg \frac{1}{\sin(\theta - \pi/2)} \\ &\ggg \frac{1}{-\cos \theta} \\ &\ggg -\sec \theta\end{aligned}$$

## Half-turn rotation

Angles  $\theta \pm \pi$  change the *sign* but keep the *trig type*. You need these essential formulas:

### 🔥 Memorize

$$\begin{aligned}\sin(\theta + \pi) &= -\sin \theta \\ \cos(\theta + \pi) &= -\cos \theta\end{aligned}$$

For  $\theta - \pi$ , just rewrite it as  $-(-\theta + \pi)$  and apply a negation identity first to peel off the outer minus.

For other trig functions, apply the definitions.

### ≡ Examples

Find  $\cos(\pi - \theta)$  like this:

$$\begin{aligned}\cos(\pi - \theta) &\ggg \cos(-\theta + \pi) \\ &\ggg -\cos(-\theta) \\ &\ggg -\cos \theta\end{aligned}$$

Of course,  $\cos(\theta - \pi) = \cos(\pi - \theta)$ . (Negate the inside.)

---

Find  $\sin(\theta - \pi)$  like this:

$$\begin{aligned}\sin(\theta - \pi) &\ggg \sin(-(-\theta + \pi)) \\ &\ggg -\sin(-\theta + \pi) \\ &\ggg - -\sin(-\theta) \\ &\ggg -\sin \theta\end{aligned}$$

Of course,  $\sin(\pi - \theta) = -\sin(\theta - \pi) = \sin \theta$ .

Find  $\tan(\pi - \theta)$  like this:

First,  $\tan(-\theta) = \frac{\sin(-\theta)}{\cos(-\theta)} = \frac{-\sin \theta}{\cos \theta} = -\tan \theta.$

Then:

$$\begin{aligned}\tan(\pi - \theta) &\ggg \tan(-\theta + \pi) \\ &\ggg \frac{\sin(-\theta + \pi)}{\cos(-\theta + \pi)} \\ &\ggg \frac{-\sin(-\theta)}{-\cos(-\theta)} \\ &\ggg \frac{\sin \theta}{-\cos \theta} \\ &\ggg -\tan \theta\end{aligned}$$

Of course,  $\tan(\theta - \pi) = -\tan(\pi - \theta) = \tan \theta.$